

7812, provide 5 and 12 V, respectively. Other 7800 series power regulators are designed for 15, 18, 20, and 24 V. The 7900 series provides negative power supply voltages in similar increments. The current capacity of the 7800 and 7900 series that come in the TO-220 style transistor packages (these can often be identified as they have no suffix or use a “T” suffix in their part number) is limited to less than 1 A. As a result, you must use them in circuits that do not draw in an excess of this amount.

2.4 The Microcontroller

A microcontroller (or a microprocessor) is a device which receives the signal given by the sensors and processes it. A microcontroller has a predefined set of conditions and corresponding actions to be taken for each of them.

If voltage given by sensor is (conditions)	Do the following (Instructions)
Less than 3 volts	Give 5 volt at the output
Equal to 3 volts	Give 6 volt at the output
Greater than 3 volts	Give 7 volt at the output

Processing a signal essentially means checking that with which condition does the input matches. For example, if the input given by the sensors to the microcontroller is 2 V, the microcontroller will immediately provide 5 volt at its output. Many vendors manufacture microcontrollers. The most popular products and their vendors are given in the following table:

Name	Manufacturer	Comments
PIC	Microchip	Huge product line that is very popular with the robot makers. The PIC16F84 has been a one of the favourites of hobbyists
AVR	Atmel	Becoming increasingly popular part with robot enthusiasts. Good free tools available for standard and MegaAVR parts.
H8	Hitachi	Mainly used in many commercial robots. Tools available on the Internet but fewer example applications than available for other devices. Not quite famous for robotic projects.
8051	Intel (originally)	One of the oldest, and thus many different varieties and part numbers available from a lot of manufacturers. The original parts are very simple but now different manufacturers have MCUs with sophisticated interfaces. Lots of examples, tools and support available on the Internet.